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Response Inhibition, Hyperactivity, and Conduct Problems Among Preschool Children

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Investigated the relation among response inhibition, hyperactivity, and conduct problems in a nonclinical sample of 115 preschool children, using 2 different types of go/no-go tasks as well as a Stroop-like task. In line with the assumption that hyperactivity is related to disinhibition, the results showed that it was the measures of response inhibition, and not other performance measures, that were related to teacher ratings of hyperactivity. There was also a significant relation between response inhibition and conduct problems. Interestingly, the correlation between response inhibition and conduct problems was not significant when partialling out the effect of hyperactivity, whereas the correlation between inhibition and hyperactivity did remain significant when controlling for conduct problems. Although the association between inhibition, hyperactivity, and conduct problems appeared to be partly different for boys and girls, these differences were not statistically significant.

The concept of behavioral disinhibition plays a central role in many models of developmental psychopathology. Although a variety of disorders have been assumed to have roots in inhibitory dysfunction, the concept has been most extensively used in the analysis of externalizing behavior problems such as attention deficit hyperactivity disorder (ADHD), conduct disorder (CD), and oppositional defiant disorder (ODD). Quay (1988) based his ideas on Gray's (1982) formulations regarding the function of two conceptual brain systems: the behavioral inhibition system and the behavioral activation system. An underactive behavioral inhibition system would assumedly result in problem behaviors associated with ADHD, whereas an underactive behavioral inhibition system, in combination with an overactive behavioral activation system, would result in CD (Daugherty, Quay, & Ramos, 1993). More recently, Barkley (1997a, 1997b, 1998) has presented an influential theoretical framework, where he postulates that ADHD should be seen primarily as a disorder of behavioral disinhibition, with links to different types of executive functions that are dependent on inhibition for their effective execution. According to Barkley (1997b), behavioral disinhibition comprises inhibition of a prepotent response, stopping of an ongoing response, and interference control, and children with ADHD are assumed to show a deficiency regarding all of these three, interrelated aspects of inhibition.

Children diagnosed with ADHD have been shown to differ from normal controls on a variety of tests of inhibition such as go/no-go tasks (e.g., Barkley, 1998; Pennington & Ozonoff, 1996) and continuous performance tasks (Barkley, Grodzinsky, & DuPaul, 1992; Grodzinsky & Diamond, 1992; see also Corkum & Siegel, 1993, for a review). Both of these types of tasks require the participants to emit a motor response as fast as possible when cued to do so and then inhibit their response when another cue is presented. However, whereas the continuous performance tasks paradigm typically involves a minority response and consequently inhibition of response to the majority of stimuli, the opposite is true for the go/no-go paradigm, and such tasks therefore better fit the requirements for tests of inhibition of a prepotent response (Barkley, 1997a). Further, a majority of the studies using tasks such as the stop-signal task and the Wisconsin Card Sort Test have shown that children with ADHD also have difficulties with inhibiting ongoing response patterns when signaled to do so (for a review, see Barkley, 1997a; Oosterlaan, Logan, & Sergeant, 1998). Regarding interference control, the third inhibitory process according to Barkley (1997b), previous studies have found that distractions outside the immediate task are not likely to differentiate between children with ADHD and normal controls, whereas distractions embedded in the tasks, such as the Stroop task (Stroop, 1935), are likely to do so (Barkley et al., 1992; Grodzinsky & Diamond, 1992; Leung & Connolly, 1996). These previous studies provide quite consistent empirical support for the assumption that children with ADHD have a deficit in inhibition.

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Behavioral disinhibition has also been associated with CD (e.g., Hurt & Naglieri, 1992; Oosterlaan et al., 1998), which raises the question of whether disinhibition is truly the hallmark of ADHD, as it does not appear to be specific to the disorder. However, because most previous studies have failed to control for comorbid diagnoses, it has been suggested that the association between disinhibition and CD might be accounted for by the large overlap with ADHD (Pennington & Ozonoff, 1996). On the other hand, it is of course also possible that the association between disinhibition and ADHD is caused by the overlap with CD. Finally, it has been argued by Lynam (1998), for example, that it is the combination of high levels of ADHD symptoms and conduct problems that is associated with poor inhibitory control, and this comorbid group has also been shown to have the most serious negative outcomes (e.g., Moffitt, 1990).

When studying hyperactivity and conduct problems, it is also important to take into consideration that, even though children diagnosed with, for example, CD, may not meet the criteria for comorbid ADHD, they may still have considerably higher levels of ADHD symptoms compared to normal controls. As a consequence, it is important to treat data dimensionally, instead of just categorically, and in that way control for comorbid symptoms at a subclinical level (Nigg, Hinshaw, Carte, & Treuting, 1998). To our knowledge, only one previous study has controlled for subclinical symptoms, and the results indicate that disinhibition is specific to ADHD and not primarily related to CD (Nigg et al., 1998). However, more research is needed within this area before any certain conclusions can be drawn.

In spite of the growing acknowledgment of the possibility that ADHD should be seen as the extreme end of a continuum of inhibitory functioning and associated behaviors, instead of representing a separate category, practically all previous studies of inhibitory functions have used children already diagnosed with ADHD. Further, because ADHD children are usually not diagnosed until they reach school, few previous studies have used samples of preschool children. In fact, all the clinical studies referred to previously have used samples of school-age children. If a positive relation between hyperactivity and inhibitory functioning can be found in normal samples as early as the preschool age, this could be seen as further support for the hypothesis that poor behavioral inhibition is the major underlying deficit in ADHD. Furthermore, if preschool children with high levels of hyperactive behavior are shown to perform poorly on tests of response inhibition, these tests might be able to improve the possibilities of identifying children who are at high risk of developing ADHD later in life.

The few previous studies that have studied behavioral inhibition in nonclinical samples of preschool children have found significant relations between inhibitory functioning and externalizing behavior problems. Nigg, Quamma, Greenberg, and Kusche (1999) found that, after correcting for autocorrelation over time for both predictor and outcome variables, there were modest effects of behavioral inhibition on behavior outcome. Furthermore, a British longitudinal study of "hard-to-manage" preschoolers found that children rated by their parents as disruptive performed more poorly on tests of inhibition (Hughes, Dunn, & White, 1998; Hughes, White, Sharpen, & Dunn, 2000). However, because the problem behavior inventories employed in these studies included both hyperactivity and conduct problems, it is not known whether it is one of these two problem types, or both of them, that are related to response inhibition.

Another issue that has seldom been addressed concerns the question of whether the relation between response inhibition and hyperactivity is the same for boys and girls. Because the sample sizes in clinical ADHD studies often are small, and the boy–girl ratio in clinical samples often range between 5:1 to 9:1 (American Psychiatric Association [APA], 1994; Barkley, 1998), either girls have been excluded from previous studies or the number of girls has been too small to conduct separate analyses for each sex. The few studies that have examined both boys and girls have mainly focused on sex differences in prevalence, severity of problems, and familial psychopathology (for a review, see Gaub & Carlson, 1997; Henker & Whalen, 1999; Whalen & Henker, 1998). In their meta-analysis of sex differences in ADHD, Gaub and Carlson concluded that although ADHD girls showed lower levels of hyperactive behavior compared to boys, they were more intellectually impaired. However, these results were found only for clinic-referred ADHD girls, which might suggest that girls with ADHD are less likely to be referred to clinics compared to boys and that ADHD girls in clinic-referred samples might not be representative for the disordered population in general (Carlson, Tamm, & Gaub, 1997). The same argument can be made regarding sex differences among children with CD. Based on the issues raised here, it has been argued that the use of population-based samples to study sex differences in ADHD, CD, and ODD is particularly appropriate (Carlson et al., 1997). Because poor response inhibition has been suggested as a fundamental mechanism behind ADHD, it should be considered important to study possible sex differences in response inhibition, as well as to study the association between response inhibition and hyperactivity using separate analyses for boys and girls.

Aims of This Study

The aim of this study was to investigate whether differences in response inhibition were related to hyperactivity, conduct problems, or both in a nonclinical sample of preschool children. Unlike previous studies, both boys and girls were included, and the effect of response inhibition on problem behaviors was studied separately for each sex. Three different computerized tests were used, all of which were similar to tests that have been shown to discriminate between ADHD children and controls, and together these tests tapped different aspects of behavioral disinhibition as defined by Barkley (1997b). Finally, because previous studies have shown that hyperactivity and conduct problems often co-occur, we wished to control for conduct problems when studying the effects of inhibitory control on hyperactivity and to control for hyperactivity when studying the effects of inhibitory control on conduct problems. In accordance with the view that disinhibition is the central impairment in ADHD and that it is specific to this type of behavior problem, it was hypothesized that (a) only the measures of disinhibition would be related to hyperactivity, and (b) a possible relation between conduct problems and disinhibition would disappear when controlling for hyperactivity.

It should be noted that this study is part of a larger longitudinal study investigating the development of problem behaviors in children ages 5 to 9. Although this study contains data that have been collected during a time period of 18 months (5 to 6½ years), the study used cross-sectional, rather than longitudinal, analyses of the data. The reason for this is that the aim of the study was to investigate the construct of response inhibition and its relation to hyperactivity and conduct problems and not how these behaviors develop over time. The fact that laboratory measures of response inhibition were collected at two different time points should, however, be regarded as an inherent strength of this study as this design limits measurement errors due to situational variation.

Method

Participants

The participants of this study were part of a larger longitudinal study conducted at Uppsala University, Sweden. The sample included 151 children (48% boys), who had been recruited from a larger randomly selected sample. A national population-based register, which includes all Swedish residents, was used for recruiting a random sample of 500 boys and 500 girls who (a) were born between January and September 1993 and (b) lived in or in the vicinity of Uppsala, Sweden (a university town and the country's fourth largest city). The sample was approached by mail, and the response rate was 70.5% after two reminders. The parents were asked if their child had any medical or psychological problems, and 2 children were excluded from the study based on the fact that they were mentally retarded.

At the age of 5 years ($M = 5.2$ years, $SD = 0.09$), 151 children from the larger, random sample were seen in the department laboratory. The subsample seen in the laboratory was selected to closely represent the distribution in the larger, random sample based on a variable screening for hyperactivity. By matching the two distributions in terms of deciles, a full variation with regard to the phenomena of interest was ascertained in the selected sample. Ratings of conduct problems were not used as a matching variable, but because of a considerable overlap between hyperactivity and conduct problems ($r = .72$ for girls and .73 for boys), matching for hyperactivity also resulted in a more or less full variation with regard to conduct problems. As might be expected from a university town, parental education was high. Only 3% of the mothers and 7% of the fathers had the 9-year compulsory school as their only schooling, 24% of the mothers and 17% of the fathers had vocational training, 24% of the mothers and 25% of the fathers had completed secondary school (12 years of schooling), and 53% of the mothers and 51% of the fathers held a college or university degree. A minority of the children, about 7%, did not have any siblings, 65% had one sibling, and the remaining children had two or more siblings.

The subsample of 151 children did not differ from the larger randomly selected sample with regard to background variables such as number of siblings, $t(697) < 1.52$, *ns*, and maternal and paternal age, both $ts(697) < 0.28$, *ns*, and education, both $\chi^2(6, N = 692) < 9.02$, *ns*. About 9 months after the visit to the department laboratory ($M = 6.0$ years, $SD = 0.22$), the children's nursery school teachers or day-care providers in the home completed questionnaires regarding behavior problems in the out-of-home setting for 124 of the children. Reasons for attrition regarding the teacher ratings were that the child did not have out-of-home care (4 children), the parents did not consent to contact with the child's nursery school (13 children), or the questionnaires were not returned despite two reminders (10 children).

When the children were about 6½ years of age ($M = 6.6$ years, $SD = 0.11$), 133 children visited the department laboratory once again.[1] Reasons for not participating in the second visit were mainly lack of time (12 children) or the fact that the family had moved out of the area (6 children). This study includes the 115 children who had complete data for both laboratory visits, as well as for the teacher ratings. The children who did not have complete data, and who were therefore not included in this study, did not differ from the larger group

[1]In Sweden, children do not start school until they are about 7 years old, and the children in this study should therefore be considered as preschoolers.

of 151 children who were recruited from the random sample with regard to parental ratings of problem behaviors (i.e., hyperactivity and conduct problems) at the time of the first visit, all ts(149) < 0.53, *ns*, or regarding errors of response inhibition on the go/no-go test administered at the first laboratory visit at age 5, $t(149) = 1.04$, *ns*. Informed, written consent from the parents, as well as verbal consent from the participating children, were obtained for each of the assessment sessions.

Measures

Go/no-go task (5 years). To test the child's ability to inhibit a prepotent response, this study included two tasks based on the go/no-go paradigm. This paradigm requires the participant to emit a simple motor response ("go") to one cue, while inhibiting this response ("no-go") in the presence of another, and has been widely used in ADHD research (e.g., Iaboni, Douglas, & Baker, 1995; Shue & Douglas, 1992; Trommer, Hoeppner, Lorber, & Armstrong, 1988). Performance on go/no-go tasks has been shown to be impaired in children with ADHD as young as 6 years old, adults with frontal lobe tumors, and experimental animals with frontal lobe lesions, suggesting that the go/no-go paradigm taps frontally based functioning that is associated with ADHD (for reviews, see Pennington & Ozenoff, 1996; Trommer et al., 1988).

In the go/no-go task administered at 5 years of age, four different stimuli (a blue square, a blue triangle, a red square, and a red triangle) were presented on a computer screen. During the first part of the task, the children were instructed to press a key ("go") when a frequent stimulus (a blue figure) appeared on the screen but to make no response ("no-go") when an infrequent stimulus (a red figure) appeared. The same stimuli were used for the second part of the task, but the children were instructed to press a key every time they saw a square and to inhibit their response every time they saw a triangle, irrespective of color. Altogether the task included 60 stimuli with a "go-rate" of 70%. Each stimulus was presented during a time period of 800 msec, followed by a response time of 1700 msec and a waiting period of 1700 msec before the next stimulus was presented. Scores derived from the task were commission errors (pressing the key when a "no-go" target was presented) and omission errors (failing to press the key when a "go" stimulus was presented). In line with Barkley's (1997b) theory of ADHD, commission errors were considered as a direct measure of inhibitory control, whereas omission errors were included to study the hypothesis that hyperactivity is primarily related to response inhibition and not to more generalized performance deficits.

Go/no-go task (6½ years). During the second visit, inhibition of a prepotent response was once again measured with a go/no-go task, this time using slightly more complex figures such as a square with a vertical or diagonal line in the middle. The children were presented with four different figures and instructed to press a key as fast as possible every time they saw one of three figures ("go"), but to inhibit their response when they saw the fourth figure ("no-go"). The figures were presented in a random order, and altogether the test included 30 stimuli, with an interval between each stimulus of 5000 msec. Scores derived from the task were commission and omission errors.

Stroop-like task. The original Stroop test (Stroop, 1935) contains words denoting colors printed in an incongruent color (e.g., the word *blue* printed in *red*). The test is meant to be a measure of interference control, as it produces a response conflict when the participant has to inhibit an overlearned response (e.g., reading the word *blue*) and instead name the color of the written word (e.g., *red*). However, because the participants have to be able to read proficiently for the test to produce an interference, the original Stroop test could not be used with the young participants in this study. Instead, a Stroop-like task using pictures was constructed. The task was based on the day–night Stroop-like task by Gerstadt, Hong, and Diamond (1994), but because previous studies using a Stroop-like task with only two stimuli, as Gerstadt and colleagues did, found that the children were performing at ceiling already by age 6 (Passler, Isaac, & Hynd, 1985), new stimuli were added and the task was computerized. In this study, the children were presented with four pairs of pictures, where the pictures in each pair were each other's opposites (day–night, large–small, boy–girl, and up–down). After making sure that the child understood what each picture represented, the child was instructed to say the opposite as fast as possible every time he or she saw a picture on the computer screen (e.g., to say "boy" every time they saw a girl). During the first part of the task, the child was presented with each picture three times in a random order, but the pairs were not mixed (i.e., the six first pictures were either a boy or a girl, the next six pictures were either large or small, and so on). During the second part, the instructions were the same and each picture was presented three times, but the eight pictures were presented in a totally random order. Each stimulus was presented during a time period of 1500 msec (1000 msec for the second part of the task), followed by a response time of 1500 msec and a waiting period of 1500 msec before the next stimulus was presented. Three different types of errors were registered: not corrected errors (naming the picture instead of saying the opposite); corrected errors (naming the picture, or starting to name the picture, and then correcting oneself); and no answer (not giving any answer). Because the measures "not corrected errors" and "no answer" could result from the fact that the child did not fully understand the task, corrected errors were seen as the most valid measure of interference control.

The Conners Teacher Rating Scale. The Conners Teacher Rating Scale is a frequently used hyperactivity measure. In this study, we used the abbreviated, 10-item version of the scale (Abbreviated Teacher Rating Scale; Conners, 1990), which has been translated and back-translated with the assistance of a bilingual person. The Swedish version of this scale has been used successfully in previous clinical studies of, for example, treatment effects among ADHD children (e.g., Gillberg et al., 1997). Factor analysis (Parker, Sitarenios, & Conners, 1999) has shown that the first six items of the scale tap hyperactivity/impulsivity (e.g., restless, impulsive, constantly moving around, failing to concentrate), whereas the last four items of the scale are considered as a measure of emotional lability (e.g., temper tantrums, cries easily). Thus, the first six items of the scale correspond very well with the criteria for ADHD according to the *DSM–IV* (APA, 1994), whereas the last four questions reflect behavior associated with hyperactivity rather than hyperactivity itself. Because we strove to keep the hyperactivity measure as pure as possible, the last four items were excluded from the scale, resulting in a six-item scale with an internal consistency, expressed as Cronbach's α, of .89. Ratings were made on a 4-point scale ranging from 0 (*does not apply at all*) to 3 (*applies very well*), and a hyperactivity score was calculated as a mean score on all six items. It should be noted that although this rating scale has often been used in clinical studies to differentiate between ADHD and comparison controls and evaluate the effect of drug treatment (Conners, 1998), the use of the term *hyperactivity* in this report does not suggest a diagnosis of ADHD.

The mean value on Conners Teacher Rating Scale was 0.50 (SD = 0.68) for the whole sample and 0.62 (SD = 0.70) and 0.40 (SD = 0.65) for boys and girls, respectively. Normative data for Swedish children are not available, but for boys, a mean value of 0.75 (SD = 0.75) was obtained in a sample of 370 Swedish boys between the ages of 6 and 8 (Nyberg, Bohlin, Berlin, & Janols, in press). When comparing the two samples, 9.3% of the boys in this study scored above the 90th percentile in the larger sample and 4.7% scored above the 95th percentile. Data for girls are, unfortunately, not available.

The Preschool Behavior Questionnaire. The Preschool Behavior Questionnaire (Behar & Stringfield, 1974; Hagekull & Bohlin, 1992) was used to measure conduct problem behavior. The Preschool Behavior Questionnaire is a preschool version of the Children's Behavior Questionnaire by Rutter Tizard, and Whitmore (1970) and has been used previously for Swedish preschoolers (see Hagekull & Bohlin, 1992, 1994, for psychometric properties of the Swedish version of this scale). The scale includes the following 10 items: fights or teases, lies, bites or kicks when angry, disobedient, destructive, often gets into fights, fails to show consideration for others, cannot share (e.g., toys) with others, blames others when things go wrong, and often has temper tantrums. Although these items are similar to items included in the *DSM–IV* (APA, 1994) to describe CD or ODD, it should be noted that, just as with hyperactivity, the use of the term *conduct problems* does not suggest a diagnosis. Cronbach's α was .89 for all 10 items. Ratings were made on a 5-point scale ranging from 1 (*does not apply at all*) to 5 (*applies very well*), and a conduct problem score was calculated as a mean score on all 10 items.

The mean value on the conduct problem scale was 1.73 (SD = 0.86) for the whole sample and 1.93 (SD = 0.93) and 1.55 (SD = 0.76) for boys and girls, respectively. This can be compared with that of another sample of 371 Swedish 6-year-old children (197 boys and 174 girls) for which a mean value of 1.90 (SD = 0.95) was obtained for the whole sample, 2.16 (SD = 1.01) for boys and 1.55 (SD = 0.76) for girls (Hammarberg & Hagekull, 2001). When comparing the two samples, 5.5% of the boys in this study scored above the 90th percentile in the larger sample and 3.6% scored above the 95th percentile. For girls, 5.0% scored above the 90th percentile and 3.3% scored above the 95th percentile.

Peabody Picture Vocabulary Test–Revised. The Peabody Picture Vocabulary Test–Revised (Dunn & Dunn, 1981) was used to assess verbal intelligence. Standard scores ranged from 66 to 137 (M = 103, SD = 15), which indicates that the scores were relatively normally distributed and ranged from about –2 to +2 standard deviations from the mean.

Procedure

The first laboratory assessment took place at the department laboratory when the children were about 5 years old. This visit took approximately 1 hr and included one test of response inhibition (go/no-go task; see the Measures section for complete descriptions of the tasks), as well as some other tasks that are reported elsewhere. The second visit, when the children were about 6½ years of age, also took place at the department laboratory and included a go/no-go task as well as a Stroop-like task. Before each laboratory visit, the child and the accompanying parent were informed about the procedure and that the session was going to be videotaped. All children and parents agreed to this procedure, and written consent forms were obtained from the parents. Between these two visits, the children's preschool teachers were contacted by mail and asked to fill out a short questionnaire, including the hyperactivity and conduct problem instruments. Before contacting the teachers, a letter in which the questionnaire was described was sent out to the participants' parents, and the parents were asked to fill out and return a written consent form.

Statistical Analyses

Pearson's product–moment correlation coefficients were used to study the relations between the different performance measures and the problem behaviors. To study possible sex differences in task performance and level of problem behaviors, *t* tests were used, and correlation coefficients between performance and problem behaviors were computed for the total sample as well as separately for boys and girls. Further, Fisher's *r* to *z* transformations were used to study differences in correlations between boys and girls. Finally, to study whether inhibitory control is specifically linked to hyperactivity, the effect of conduct problems was partialled out when computing the correlation between inibition and hyperactivity, and the effect of hyperactivity was partialled out when studying the correlation between inhibition and conduct problems.

The analyses in this study were reanalyzed, controlling for verbal intelligence, but because no conclusions were changed, these results are not reported. Furthermore, the practice of controlling for intelligence has been questioned as performance on IQ tests may be directly related to deficient executive functioning (Barkley, 1997a). Previous studies of response inhibition have seldom reported the results both with and without controlling for intelligence, but when this has been done, intelligence has been shown to have little effect (e.g., Iaboni et al., 1995; Oosterlaan & Sergeant, 1996; Schachar & Logan, 1990; Schachar, Tannock, Marriott, & Logan, 1995).

Results

Regarding inhibition of a prepotent response, the results showed a significant relation between commission errors and problem behaviors (see Table 1; for descriptive data on the dependent and independent variables, see Table 3). Children who were rated as high in hyperactivity and conduct problems were more likely to fail to inhibit a response when a target was presented. This effect was found for the go/no-go task administered at 5 years, as well as for the similar task administered at 6½ years of age. However, there were no significant relations between omission errors and any of the problem behaviors. Regarding interference control and its relation with hyperactivity and conduct problems, there was a significant relation between corrected errors and problem behavior. However, neither hyperactivity nor conduct problems was related to the other types of errors on the Stroop-like task (see Table 1). In other words, just saying the wrong word, or not answering at all, was not related to problem behavior, whereas saying the wrong word or starting to say the wrong word and then correcting one's response was shown to be related to higher levels of both hyperactivity and conduct problems.

Because the three variables that were significantly related to problem behavior, that is, commission errors on the two go/no-go tasks and corrected errors on the Stroop-like task, were meant to capture the same construct, an aggregated measure was created using standardized values. The correlations between these three measures, which were obtained over a time period of 18 months, ranged from .24 to .34 (all *p*s $< .01$), and the aggregated measure of response inhibition was shown to be significantly correlated with both hyperactivity and conduct problems. As can be seen in Table 2, response inhibition was shown to explain about 19% of the variance in hyperactivity and about 13% of the variance in conduct problems.

Table 1. *Pearson's Product Moment Correlations Between the Three Laboratory Tasks and Teacher Ratings of Hyperactivity and Conduct Problems*

	Hyperactivity	Conduct Problems
Go/no-go task (5 years)		
Commission errors	.39***	.34***
Omission errors	.08	.11
Go/no-go task (6½ years)		
Commission errors	.25**	.25**
Omission errors	.02	.02
Stroop-like task		
Corrected errors	.30**	.19*
Not corrected errors	.03	.03
No answer	.11	.06

$^*p < .05$. $^{**}p < .01$. $^{***}p < .001$.

Sex Differences

Regarding sex differences in level of problem behavior, boys were rated as having significantly higher levels of conduct problems compared to girls, whereas the difference in hyperactivity did not reach significance (see Table 3). Regarding the performance measures, boys were shown to make a larger number of errors that could be attributed to poor response inhibition (i.e., commission errors on the two go/no-go tasks and corrected errors on the Stroop-like test), whereas there

Table 2. *Pearson's Product–Moment Correlations Between Aggregated Measure of Response Inhibition and Teacher Ratings for the Total Sample and Each Sex*

	Response Inhibition		
	Total Sample	Boys	Girls
Hyperactivity	.43***	.47***	.33**
Conduct problems	.36***	.28*	.32*

$^*p < .05$. $^{**}p < .01$. $^{***}p < .001$.

Table 3. *Means for Boys and Girls on Teacher Ratings and the Three Laboratory Tasks*

	Boys (n = 54)		Girls (n = 61)		
	M	*SD*	*M*	*SD*	*t*
Hyperactivity	0.62	0.70	0.40	0.65	1.75
Conduct problems	1.95	0.93	1.56	0.76	2.40*
Go/no-go task (5 years)					
Commission errors	4.22	3.25	2.22	2.17	3.82***
Omission errors	4.33	4.24	3.12	3.31	1.71
Go/No-go task (6½ years)					
Commission errors	4.01	1.92	2.76	2.04	3.36***
Omission errors	2.36	3.15	3.11	4.40	1.06
Stroop-like task					
Corrected errors	5.46	3.93	3.23	2.52	3.57***
Not corrected errors	1.93	4.27	1.01	2.59	1.36
No answer	1.31	2.82	0.62	1.50	1.61

*$p < .05$. **$p < .01$. ***$p < .001$.

were no significant sex differences regarding the other measures of performance.

To study possible sex differences, and to make sure that the significant relations between performance and problem behaviors could not be explained as merely an effect of sex, correlations were conducted between the disinhibition measure and the ratings of problem behaviors for each sex separately. As can be seen in Table 2, there was a significant relation between inhibition and both hyperactivity and conduct problems for boys as well as girls. Although the relation between response inhibition and hyperactivity was somewhat stronger for boys compared to girls, this sex difference was not statistically significant (z = .89, *ns*).

Is Disinhibition Specifically Related to Hyperactivity?

Although the results showed a significant relation between disinhibition and conduct problems, it was hypothesized that this could be an effect of the large overlap between conduct problems and hyperactivity. To study if disinhibition is unique to hyperactivity, the effect of hyperactivity was controlled for when computing correlations between disinhibition and conduct problems, and the effect of conduct problems was controlled for when studying the association between disinhibition and hyperactivity. When studying the sample as a whole, the results showed that the correlation between disinhibition and hyperactivity was shown to be significant when controlling for conduct problems (r = .26, $p < .01$), whereas the association between disinhibition and conduct problems did not remain significant when controlling for hyperactivity (r = .07, *ns*).

In the analyses of each sex separately, it was found that for boys the results were consistent with the results for the whole sample, as the relation between disinhibition and hyperactivity was significant when controlling for conduct problems (r = .39, $p < .01$), whereas the relation between disinhibition and conduct problems did not remain significant when controlling for hyperactivity (r = –.09, *ns*). On the other hand, when conducting the same analyses for girls, the results showed that neither the correlation between disinhibition and hyperactivity (r = .15, *ns*) nor the correlation between disinhibition and conduct problems (r = .13, *ns*) was significant when controlling for comorbid conduct problems or hyperactivity. Although these results indicate that there might be sex differences in the association between inhibition, conduct problems, and hyperactivity, further analyses of these correlation coefficients showed that, when controlling for conduct problems or hyperactivity, neither the correlation between inhibition and hyperactivity (z = 1.39, *ns*) nor the correlation between inhibition and conduct problems (z = 1.18, *ns*), differed significantly between boys and girls.

Finally, the possibility that it is the combination of hyperactivity and conduct problems that is associated with response inhibition was tested. The results showed that hyperactivity and conduct problems explained almost 19% of the variance in inhibition, R^2 = .188, $F(2, 112) = 12.97$, $p < .001$, but when entering the interaction term into a regression model, the change in R^2 was only .008, $F(3, 111) = 1.03$, *ns*. When conducting the same analyses separately for each sex, the results showed that the change in R^2 was only .008, $F(3, 50) = 0.48$, *ns,* for boys and .002, $F(3, 57) = 0.14$, *ns,* for girls.

Discussion

The results showed that commission errors on the two go/no-go tasks and corrected errors on the Stroop-like test were related to hyperactivity and conduct problems. In other words, only the measures of disinhibition were related to problem behavior and not the other measures of performance. Further, boys were shown to make more errors of inhibition compared to girls, but no sex-differences were found regarding the other measures. The aggregated measures of disinhibition were shown to explain 19% of the variance in hyperactive behavior and about 13% in conduct problems. The results also showed that the correlation between disinhibition and conduct problems was no longer significant when controlling for hyperactivity, whereas the correlation between disinhibition and hyperactivity did remain significant when controlling for conduct problems. The interaction of hyperactivity and conduct problems was not shown to be related to inhibition.

To our knowledge, the relation between disinhibition and hyperactivity has not been studied previously in nonclinical samples of either preschool or school-age children. However, our findings are consistent with previous clinical studies, which have shown significant differences between ADHD children and controls on tests of response inhibition (e.g., Barkley, 1997a, 1997b; Douglas, 1988; Pennington & Ozonoff, 1996; Shue & Douglas, 1992), as well as with the population-based studies that have shown a significant relation between disinhibition and general disruptive behavior (Hughes et al., 2000; Nigg et al., 1999). Taken together, the results of this study and previous clinical and population-based studies lend consistent support to the theoretical notion of a deficient behavioral inhibition system among hyperactive children.

Because disruptive behavior includes conduct problems as well as hyperactivity, the population-based studies mentioned previously not only suggest a relation between response inhibition and hyperactivity, but also a relation between inhibition and conduct problems. This is in line with the results of this study, which did show such as relation, as well as with similar findings obtained in clinical settings where deficits in response inhibition have been found among children with CD (Hurt & Naglieri, 1992; Oosterlaan et al., 1998; Quay, 1997). However, other studies have failed to find a relation between response inhibition and CD (Schachar & Tannock, 1995; Schachar, Tannock, & Logan, 1993), and it has been suggested that the association between response inhibition and CD is caused by the large overlap between ADHD and CD (Pennington & Ozonoff, 1996). Because children with CD often show elevated levels of ADHD symptoms without meeting all criteria for a diagnosis, it has been emphasized that the only way to control for subclinical symptoms is to treat these symptoms as dimensions and remove all of their influence by statistical means (Nigg et al., 1998).

This study did control for subclinical symptoms, and, interestingly, the results showed that the correlation between response inhibition and conduct problems was no longer significant when controlling for hyperactivity, whereas the correlation between disinhibition and hyperactivity did remain significant when controlling for conduct problems. Further, the results did not show hyperactivity and conduct problems to interact in relation to disinhibition. These findings indicate that the association between inhibition and conduct problems was caused by the large overlap between conduct problems and hyperactivity, and it provides further support for the role of response inhibition in understanding the mechanisms behind hyperactivity.

It should, however, be noted that inhibition was only shown to explain about one fifth of the variance in hyperactivity, which could be regarded as rather modest for a variable taken to represent the core deficit in ADHD (Barkley, 1997b). Further studies with clinical as well as nonclinical samples are needed to estimate the true amount of variance explained, taking into account not only attenuation due to measurement error, but also age of the child and type of inhibition measure. A significant but modest association could also suggest that response inhibition is only one manifestation of a more general regulatory control problem. Such a view has been presented by Douglas (1999), who claimed that children with ADHD fail to allocate adequate attention or effort to meet task demands (i.e., they have both an inhibitory and an activating problem). Similar ideas have also been presented by Sergeant, Oosterlaan, and van der Meere (1999), who believe that children with ADHD have deficiencies in the mechanisms supplying the energetic resources for information processing.

Regarding sex differences, the result of this study showed that the association between response inhibition and hyperactivity did not remain significant for girls when conduct problems were controlled for in the analysis. Thus, for girls, the overlap between hyperactivity and conduct problems to a large extent also overlapped with inhibition. This finding is interesting, and even though the correlations for boys and girls did not differ significantly from each other, these results could be taken as an indication that it might be premature to conclude that inhibition, hyperactivity, and conduct problems are similarly associated across the sexes. Because previous studies using nonclinical samples have not conducted separate analyses for boys and girls, further studies are needed to elucidate the issue of sex differences in this context.

Another finding from this study relates to the particular tests used. Especially the go/no-go task administered at 5 years of age, but also the other two tasks, showed a relatively large variability between individuals, which suggests that they are well adapted for studying response inhibition among preschool children. Barkley (1997a) suggested that the poor response inhibition among ADHD children should be seen as a delay rather than a deficit, and it is therefore very important to choose a task that is appropriate for the particular age group studied. The tasks used in this study, especially the go/no-go task used at age 5 and the Stroop-like task, appeared to be easy enough so that some children could perform well on them but still difficult enough so that other children had not yet learned to master them. Regarding the go/no-go task administered at age 6½, this task might have been a little too difficult, because the children in general made about the same amount of errors on this go/no-go task, despite the fact that the number of stimuli was half as many as the one administered at age 5. This fact, as well as the fact that the three tasks tapped different aspects of response inhibition, might explain the modest stability for the different measures used in this study.

Another point that should be emphasized regarding the particular tasks used in this study is that, in general, boys did not perform as well as girls. Sex differences regarding errors on go/no-go tasks or the Stroop-task have not been found using clinical samples of ADHD children (Nydén, Hjelmquist, & Gillberg, 2000). However, because many more boys compared to girls are diagnosed with ADHD (Barkley, 1998), and deficient inhibition has been suggested as the cause of the disorder (Barkley, 1997b), one would expect sex differences in disinhibition using a population-based sample. The observed sex differences in performance may, in fact, be taken as an indication that the measures of disinhibition used in this study could be useful when trying to identify children at high risk of developing clinically relevant ADHD symptoms later in childhood.

In summary, it should be noted that although many previous clinical studies have investigated response inhibition in relation to different psychiatric disorders such as ADHD and CD, this study provides one of the first insights into how response inhibition is related to hyperactivity and conduct problems in a nonclinical sample. Further, by using a dimensional rather than a categorical approach, we were able to control for symptoms of hyperactivity and conduct problems at the subclinical level, thereby providing a better understanding of the overlap between these two types of problem behaviors and how they relate to response inhibition. It should be considered important for future studies to investigate how response inhibition develops over time and whether it is possible to improve early identification of children at risk of developing ADHD by adding laboratory measures of response inhibition to information about other risk factors.

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